

- 1 A flower shop has a room that has an air-conditioning system. It is used to keep the room at a constant temperature for storing flowers.

The owner uses a keypad to set the temperature for the room.

- (a) Tick (✓) **one** box to show whether a keypad is an example of an input, output, process or storage device.

- | | |
|------------------|--------------------------|
| A input | ☐ |
| B output | ☐ |
| C process | ☐ |
| D storage | ☐ |

[1]

- (b) The owner of the shop enters a denary number to set the temperature for the room.

The denary number is converted to a binary number to be processed. The binary number is stored in a register in the air conditioning system.

- (i) Give the binary number that would be stored when the denary number 24 is entered.

..... [1]

- (ii) Give the binary number that would be stored when the denary number 101 is entered.

..... [1]

- (iii) The binary number 00010011 is stored in the register.

Give the denary number that has been entered for this binary number to be stored.

..... [1]

- (iv) The binary number 00100010 is stored in the register.

Give the denary number that has been entered for this binary number to be stored.

..... [1]

Working space

.....

.....

.....

.....

.....

.....

.....

